ARGUS Project

Mission Definition, Payload Description

ARGUS project – November 2026

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1 Mission Definition

1.1 Studied object / functional mission target

Definitive choice: space debris in low Earth orbit, a small object (about less than 1 kg, aluminium surface) to be tracked and deflected by a CubeSat.

1.2 Scientific / technological question or aim

Definitive choice: demonstrate that a 6U CubeSat can detect and track a small debris at short range with a LiDAR, and change its trajectory by laser ablation (target : small ΔV).

1.3 Main payload

The main payload of the Argus mission is the debris, which we aim to eject. The shape of the debris has not been finalized, but it will be a small object weighing less than 1 kg with an aluminium surface. The second main payload is the laser, a Class 4 laser with a wavelength of 1064 nm.

1.4 Secondary payloads (if any)

The laser will be guided by a LiDAR (secondary payload) operating at a wavelength of 1550 nm, equipped with a SPAD detector and a MEMS micro-mirror. The second secondary payload is a camera with a machine learning algorithm to detect the debris and estimate its orientation. Thanks to these two instruments, we can track the debris and fire the laser at it.

1.5 Experts contacted

Ratio of contacted experts: [**TO DO: X**] % ([**TO DO: X**] experts contacted out of [**TO DO: Y**] experts initially identified).
[**TO DO: table of experts (name, field, topic, contacted or not) – to be defined once the other subsystem is discussed**] Vincent robert for space mecanique if u have someone for laser ablation and ml and camera, please contact him

2 Payload Description

Sizing a LASER-ablation deorbiting system involves a direct physical trade-off: the LASER must not destroy the debris through excessive heat input, but must vaporise a very thin layer of its surface to create a propulsive plasma that slows the moving debris. Thus, modifying the orbit of an object (debris) relies on converting the optical energy deposited by laser pulses into the mechanical energy of the plasma jet.

2.1 LASER Ablation Propulsion

When a high-intensity LASER beam strikes the surface of a material, the optical energy is absorbed instantaneously. The temperature of the surface layer rises so rapidly that it undergoes sudden sublimation and becomes a highly energetic plasma. This jet of matter is expelled at very high speeds. Under the physical law of conservation of matter, the ejection of matter generates a mechanical recoil force on the target object [1].

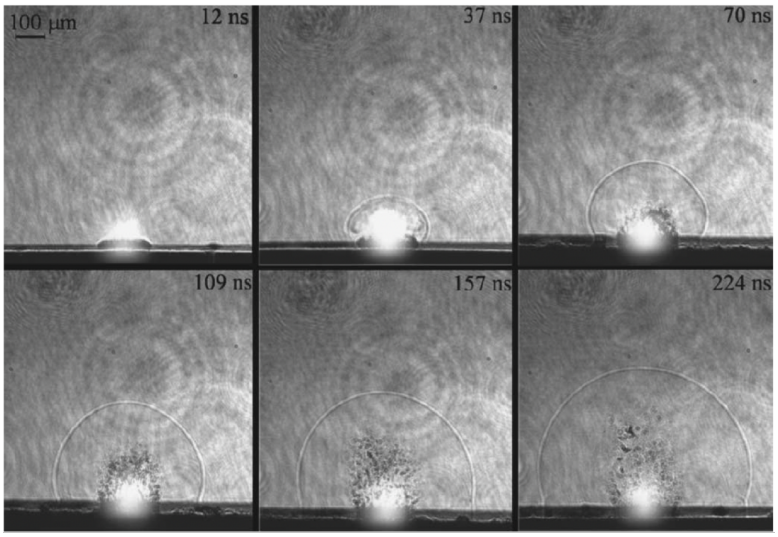


Figure 1: Images of the ablation propulsion process at different times during the laser–aluminium interaction for $\lambda = 1064nm$ and $\tau = 5ns$. The propagation of the shock wave over time can be observed. [1]

2.2 High-Power LASER Characteristics

In this section, we describe the characteristics of our onboard high-power LASER. Selecting the high-power LASER is crucial: if its power or energy is unsuitable, it will probably not be possible to initiate the micropropulsion process at the surface of our target (debris).

1. **Choice of LASER type:**
   We choose a pulsed LASER rather than a continuous-wave laser. The factors determining this choice are summarised in the table below

Table 1: Comparison between continuous-wave and pulsed lasers.

Characteristic	**Continuous-wave laser**	**Pulsed laser**
Effect on debris	Thermal melting	Plasma ablation (mechanical rocket effect)
Peak power	Equal to average power	Several megawatts during the pulse
Generated thrust	Almost zero at this power level	Optimal and directional
Risks	High risk of breaking up the debris	Safe, targeted surface action

2. **Required LASER parameters:**
The wavelength of our high-power LASER is set to λ =1064 nm in the infrared, with a pulse duration of **nanoseconds (ns)**. This choice is primarily based on the absorption characteristics of the material making up our debris, namely aluminium (Al).

For this CubeSat project using a $6U$ platform firing a LASER at short range, the optimal average power required for our small aluminium debris object, measuring 1 cm to 10 cm and weighing $1kg$ is approximately $1000W$. The choice of this operating power is deliberate and results from a specific trade-off validated by orbital mechanics and the physics of laser–matter interaction. The section on project feasibility discusses in detail the importance of reaching this optimal LASER ablation power.

Table 2: Ablation laser parameters.

Parameter	**Value**
Wavelength (λ)	1064 nm (infrared)
Operating mode	Pulsed
Pulse duration (τ)	Nanosecond (ns)
Average power (P_{moy})	1000 W (1 kW)
Repetition rate (f)	3000 Hz (2 kHz)
Pulse energy (E_p)	0,5 J
Peak power ($P_{\text{crête}}$)	100 MW
Ablation threshold fluence	1 J cm^{-2} to 2 J cm^{-2}
Optimal coupling fluence	5 J cm^{-2}

2.3 LiDAR for Detection

To meet the imposed dimensional constraints, we favour a micro-LiDAR over a conventional LiDAR, which would entail higher mass and energy demands. The micro-LiDAR provides optical metrology of the debris: detection, relative position and relative velocity [2]. It consists of three main elements:

- The emission source, which in our case could be a VCSEL (Vertical Cavity Surface Emitting LASER). These are arrays of miniature semiconductor LASERs. The table below summarises the parameters we consider optimal for defining the LASER source of our micro-LiDAR.

Table 3: Micro-LiDAR laser parameters [3].

Parameter	**Specified value**	**Purpose in space**
Wavelength	1550 nm	Prevents blinding by the ablation laser (1064 nm)
Source type	Pulsed laser diode (InGaAs)	Robustness, extreme compactness
Pulse duration	3 ns	Millimetre-level ranging accuracy
Frequency	20 kHz	Smooth real-time target tracking
Pulse energy	10 µJ	Completely harmless to the debris
Electrical power consumption	$\approx$ 12 W	Fully suited to the constraints of a 6U CubeSat

- An ultrasensitive SPAD (Single Photon Avalanche Diode) detector. These detectors can detect a single returning photon. The table below summarises the parameters we consider optimal for selecting a suitable detector for our micro-LiDAR [4].

Table 4: Micro-LiDAR SPAD detector parameters [5].

Parameter	**Specified value**	**Purpose in space**
Detector type	SPAD InGaAs/InP	Sensitive at 1550 nm, single-photon detection
Detection efficiency	$\approx$ 20 % at 1550 nm	Detect the weak signal backscattered by the debris
Timing jitter (*jitter*)	$\approx$ 200 ps	Range resolution of a few centimetres
Dark count rate	$\approx$ 10 kHz (cooled detector)	Limit noise to maintain a good signal-to-noise ratio
Operating mode	Gated (*gated*), synchronised with the pulse	Reduce noise and power consumption

- MEMS (Micro-Electro-Mecanical System) micromirrors. These oscillating micromirrors are etched into silicon to steer the beam without heavy mechanical components. The table below summarises the parameters we consider optimal for selecting the mirror assembly for our micro-LiDAR [6].

Table 5: Micro-LiDAR oscillating MEMS micromirror parameters [7].

Parameter	**Specified value**	**Purpose in space**
Scanning mechanism	Resonant MEMS micromirror	Scene scanning without bulky mechanical components
Mirror diameter	$\approx 1\,\mathrm{mm}$	Trade-off between optical aperture and resonant frequency
Optical scan angle	$\pm 10^{\circ}$	Cover the field of view required for target acquisition
Resonant frequency	$\approx 1\,\mathrm{kHz}$	Scan rate compatible with real-time tracking
Electrical power consumption	Less than 1 W	Compatible with the power budget of a 6U CubeSat

3 System Architecture and Diagrams

3.1 Data Management Plan – functional diagram V1



Figure 2: Data Management Plan – functional diagram V1.

Legend. The diagram is divided into three phases: *search of the debris* (detection and localisation), *calculation after ablation* (check of the laser effect and transmission), and *analysis of the data on ground*. Each box is a step of the chain, and the small text next to it gives the equipment performing the step

(onboard computer, lidar, camera, communication board, antenna, ground station, ground computer, or researcher). The labels on the arrows give the type of data exchanged: *picture* (raw camera image), *position octet* (debris position encoded in bytes), *octet* (binary data such as measurements and deviation), *frequency* (radio signal between the antenna and the ground station), and *text* (final publication).

3.2 Satellite platform – functional diagram V1

[TO DO: insert the figure (see the "Examples" tab)look on miro]

3.3 Satellite system model V1 (IDM-CIC)

[TO DO: insert the screenshot or the link to the IDM-CIC model]

4 Mission Analysis V1

[TO DO: orbit and mission phases – to be discussed with the other subsystem] [TO DO: maybe add some seach on procurreezy and vector ifyou have sometjing interresing]

5 Mission Report Summary V1

[TO DO: write the summary following the "Leads for Report Summary" tab: context, objectives, payloads, architecture, orbit and phases, main risks, schedule]

6 Risk Analysis

6.1 Risks related to the power LASER

Risk	Cause	Consequence	Level	Prevention
Power and heat too high for a 6U	1 kW optical with 35 % efficiency requires about 2.9 kW electrical, including nearly 1.9 kW of heat	Difficult to power and cool the system	High	Reduce P_{avg}, favour short shots on battery, perform a thermal study
Poor pointing	3 mm spot at 50 m, i.e. about 60 μrad, very demanding for a CubeSat	Shot misses the debris, no ablation	High	Fine pointing mirror guided by the LiDAR
Debris drifts away during firing	With $F = 30$ mN on 1 kg, the debris moves away by about 150 m in 100 s	Impossible to fire 2433 s continuously to reach 73 m/s	High	Short shots, debris tracking, aim for a smaller ΔV
Ablation efficiency overestimated	$C_m = 30$ μN·s/J assumed constant	Lower force and longer firing time	Medium	Take a margin, perform ground tests
Damaged optics	100 MW peak pulses, aluminium vapour deposition	Power loss, damaged optics	Medium	Margin on damage threshold, protective window
Debris fragmentation	Fluence too high or tumbling debris	New debris (Kessler syndrome)	High	Control the fluence, stop firing in case of anomaly
Unintended shot or reflection	Highly reflective aluminium surface	Dazzling or damaging another satellite	Medium	Shutter, firing authorized only if the target is locked

Risk	Cause	Consequence	Level	Prevention
Hazard to people during tests	Class 4 laser at 1064 nm	Eye or skin injury	High	Suitable goggles, closed room, laser safety officer
Source failure	Ageing of the pump diodes	Loss of the mission	Medium	Operating margin, power monitoring

6.2 Risks related to the LiDAR

Risk	Cause	Consequence	Level	Prevention
Millimetre accuracy not reached	3 ns pulse and 200 ps SPAD jitter, i.e. about 3 cm	Position less accurate than expected	Medium	Announce centimetre accuracy, justify millimetre level by averaging
Return signal too weak	Small debris, SPAD efficiency of only 20 %, aluminium surface	Debris not detected	Medium	Perform a link budget, adjust energy or frequency
Sensor dazzling	Ablation laser, plasma, Sun or Earth in the field of view	Blind detection	Medium	1550 nm filter, gated operation, shutter
Loss of debris tracking	Small tumbling target, field limited to $\pm 10°$	Firing impossible, target lost	High	Reacquisition algorithm, adapted field of view
Sensor degraded by radiation	Increasing dark noise (10 kHz at start)	False detections	Medium	Cooling, shielding, noise monitoring
Fragile MEMS micro-mirror	Vibration effects at launch	Mirror stuck or shifted	Medium	Launch lock, vibration and thermal tests
Synchronization and onboard computing	Offset between emission and detection, limited CubeSat computer	Wrong or too slow measurements	Medium	Common clock, lightweight algorithms
LiDAR eye safety	10 μJ beam at 20 kHz	Low hazard during tests	Low	Check the standard, limit access

6.3 Risks related to the camera and Machine Learning

Risk	Cause	Consequence	Level	Prevention
Missed or false debris detection	Small target, stars or reflections mistaken for debris	Target not acquired or wrong object tracked	High	Train with and without debris, check detection over several images
Poor image quality	Shadows, strong reflections or motion blur	Debris features cannot be located reliably	Medium	Adjust exposure, test different lighting conditions, reject unusable images
Training images not representative	Mostly simulated images, limited real views of the debris	Good ground results but poor performance in flight	High	Add real images, vary lighting and rotation, keep a separate test set

Risk	Cause	Consequence	Level	Prevention
Debris too small in the image	Camera sized for 2–10 m, target farther away during firing	Detection possible but orientation cannot be estimated	High	Agree on a common operating distance, check target size in pixels
Wrong distance or orientation estimate	Symmetric debris, poorly located features or camera calibration error	Incorrect tracking and pointing data	High	Calibrate the camera, check the 3D model, compare distance with the LiDAR
Loss of visual tracking	Tumbling debris or target leaving the predicted image region	Position estimate no longer valid, firing interrupted	High	Restart full-image detection, inhibit firing until tracking is confirmed
Onboard processing too slow	Two neural networks and tracking running on limited hardware	Two observations per second not reached, delayed pointing data	Medium	Test the complete chain on Q8S and XE2, simplify the models if needed
Model deployment failure	Network operations unsupported on XE2 or reduced numerical precision	Model fails to run or gives less accurate results	Medium	Compile both networks early, compare onboard results with ground results
Camera and LiDAR data out of sync	Different timestamps, transfer delays or inconsistent reference frames	Measurements refer to different target positions	High	Use a common clock, define reference frames, test the complete sensor interface

6.4 Risks related to the whole LiDAR + LASER payload

Risk	Cause	Consequence	Level	Prevention
Payload too heavy or power-hungry for a 6U	Laser, cooling, battery, LiDAR (about 12 W) and optics to accommodate	Mission not feasible in 6U	High	Mass, volume and power budget with the other teams
LiDAR / laser misalignment	Launch vibrations, temperature variations	Laser does not aim where the LiDAR measures	High	Common rigid structure, in-orbit calibration
Firing software bug	Error in the logic between detection and firing	Unintended shot or lock-up	High	Simple logic, loop testing
Electromagnetic interference	Pulsed laser power supply close to the sensor and the onboard computer	Disturbed measurements, reboots	Medium	Shielding, careful cabling, EMC tests
Shocks, vibrations and vacuum	Launch and orbital conditions	Misaligned or broken parts	Medium	Vibration, thermal and vacuum tests
Launcher safety requirements	Class 4 laser and battery on board	Integration refused	High	Laser disabled until separation, early dialogue with the launcher

Risk	Cause	Consequence	Level	Prevention
Poor debris ejection	Velocity of 0.1 to 0.5 m/s poorly controlled	Debris returning towards the CubeSat	Medium	Simulate the relative trajectory before flight
Loss of communication with the ground	Firing outside the contact windows	Uncontrolled firing	Medium	Firing inhibited by default, safe autonomous sequence
Laser cost	Space-qualified 1 kW pulsed source beyond a student budget	Project blocked	High	Get quotes from suppliers, plan a lower-power demonstrator

6.5 Human and organizational risks (related to the subsystem)

Risk	Cause	Consequence	Level	Prevention
Unnoticed calculation errors	Different values in different places of the document	Wrong results in the report	Medium	Single calculation file, review by a colleague
Limited skills	Little experience in laser, optics and laser safety	Poor technical choices, risk during tests	High	Training, advice from the supervisor or an expert
Poor communication between teams	Interfaces with power, thermal, structure and attitude not defined	Incompatibilities discovered too late	Medium	Regular meetings, interface document
Optimism on assumptions	Values chosen so that the calculation works (1 kW, 73 m/s)	Reality may be different	Medium	External review, explicit margins
Handling error during tests	Laser or high voltage without protocol	Personal injury	High	Written protocol, work in pairs, emergency stop

6.6 Risk analysis for the participants

Probability / severity, each rated out of 5 (5 = maximum).

Problem	Prob. / Severity	Solution
Loss or breakage of a computer	2/4	Backup device, handwriting
Long-term illness	3/2	Share the work, work more after recovery
Inability to work	2/5	Scale the project down, increase each person's workload
Exclusion from the master's programme	1/5	Same as above
Work overload (other courses, exams, internship)	3/4	Rely on a good schedule
Incompatible availabilities between members	4/3	Video meetings and/or state availabilities in advance, read the meeting minutes afterwards

Problem	**Prob. / Severity**	**Solution**
Poor workload estimation	3/3	Redistribute the work among us, report it in a meeting or to the coordinator
Working on the same parts	4/4	Define the tasks with the WBS
Timing problem for deliverables	3/5	Respect the milestones set at the beginning of the year